



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment: 9 Name: Dhanish Ladwani PRN: 24070521020

1. Experiment Title:

Implement localStorage and sessionStorage to save user preferences like theme settings.

2. Software/Tools Required:

Google Chrome (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS, Javascript

3. Experiment Program Code:

Experiment 9.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Theme Preferences</title>
  <link rel="stylesheet" href="Experiment 9.css">
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display=swap"
rel="stylesheet">
</head>
<body>
  <div class="container">
    <h2>Theme Preferences</h2>
    <p class="subtitle">Using localStorage and sessionStorage to save user preferences.</p>

    <div class="button-group">
      <button id="lightBtn" onclick="setTheme('light')">Light Theme</button>
      <button id="darkBtn" onclick="setTheme('dark')">Dark Theme</button>
      <button id="clearBtn" onclick="clearPreference()">Clear Preference</button>
    </div>

    <footer>
      Made by Dhanish Ladwani PRN - 24070521020
    </footer>
  </div>

  <script src="Experiment 9.js"></script>
</body>
</html>
```

Experiment 9.css

```
* {
  box-sizing: border-box;
}

body {
  background-color: whitesmoke;
  color: #333;
  font-family: 'Inter', sans-serif;
  margin: 0;
  padding: 40px 20px;
```

```

display: flex;
justify-content: center;
align-items: center;
min-height: 100vh;
transition: background-color 0.3s ease, color 0.3s ease;
}

body.dark {
background-color: #1e1e2e;
color: #cdd6f4;
}

.container {
width: 100%;
max-width: 500px;
background-color: white;
padding: 35px 30px;
border: 1px solid lightgray;
border-radius: 12px;
box-shadow: 0 4px 15px lightgray;
text-align: center;
transition: background-color 0.3s ease, border-color 0.3s ease, box-shadow 0.3s ease;
}

body.dark .container {
background-color: #252739;
border-color: #3e4259;
box-shadow: 0 4px 15px rgba(0, 0, 0, 0.4);
}

h2 {
color: darkblue;
margin-top: 0;
margin-bottom: 12px;
font-size: 24px;
transition: color 0.3s ease;
}

body.dark h2 {
color: #89b4fa;
}

.subtitle {
font-size: 14px;
color: #555;
margin-bottom: 24px;
line-height: 1.5;
transition: color 0.3s ease;
}

body.dark .subtitle {
color: #a6adc8;
}

.button-group {
display: flex;
justify-content: center;
gap: 10px;
flex-wrap: wrap;
margin-bottom: 10px;
}

button {
padding: 10px 18px;
background-color: royalblue;
color: white;
border: none;
border-radius: 6px;
font-size: 14px;

```

```

    font-weight: 600;
    cursor: pointer;
    transition: background-color 0.2s, transform 0.1s ease;
}

button:hover {
    background-color: blue;
}

button:active {
    transform: scale(0.98);
}

#clearBtn {
    background-color: #6c757d;
}

#clearBtn:hover {
    background-color: #5a6268;
}

footer {
    text-align: center;
    margin-top: 30px;
    font-size: 14px;
    color: gray;
    border-top: 1px solid lightgray;
    padding-top: 15px;
    transition: border-color 0.3s ease, color 0.3s ease;
}

body.dark footer {
    color: #a6adc8;
    border-top-color: #3e4259;
}

```

Experiment 9.js

```

function setTheme(theme) {
    if (theme === 'dark') {
        document.body.classList.add('dark');
    } else {
        document.body.classList.remove('dark');
    }

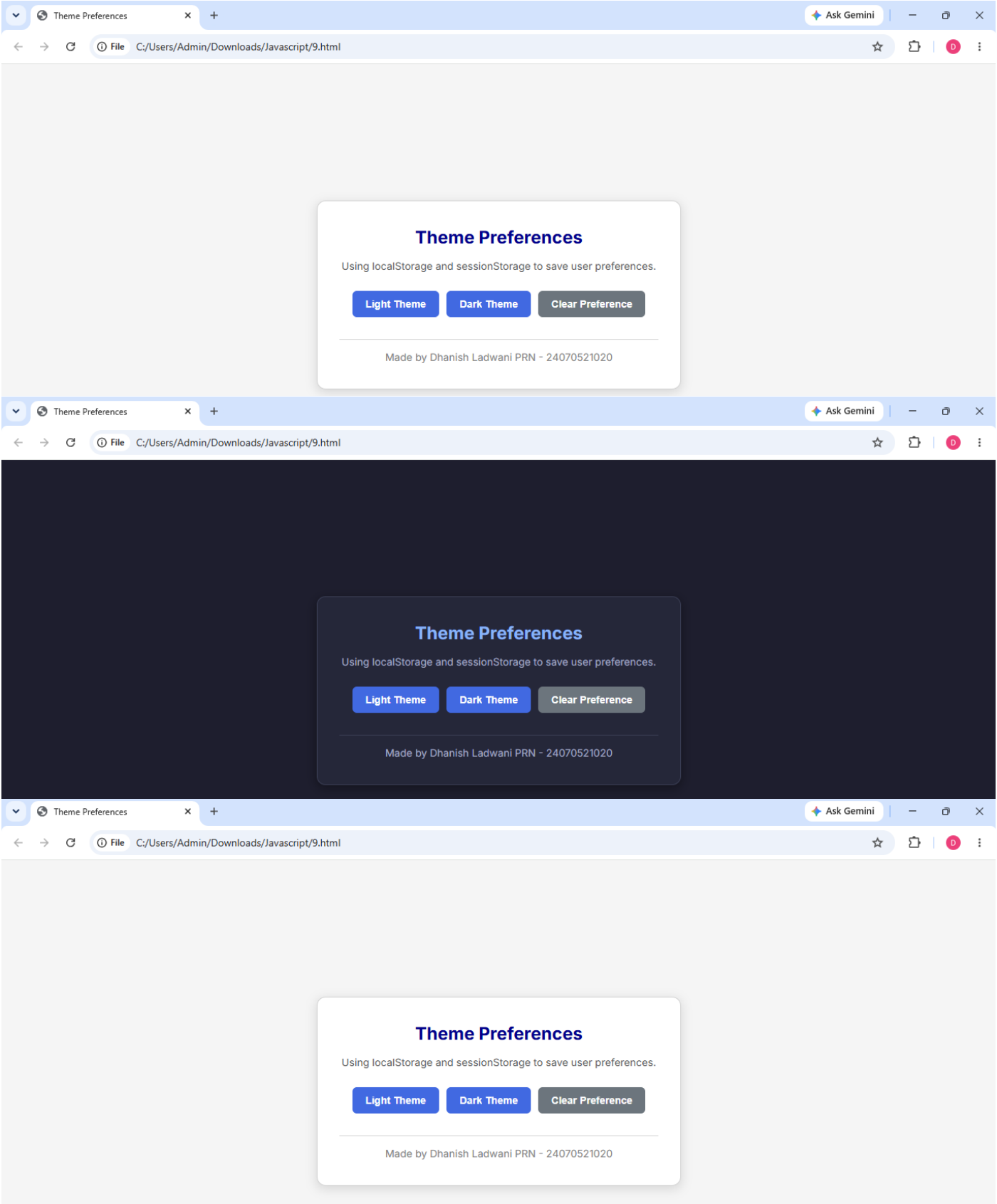
    localStorage.setItem('theme', theme);
    sessionStorage.setItem('sessionTheme', theme);
}

function clearPreference() {
    localStorage.removeItem('theme');
    sessionStorage.removeItem('sessionTheme');
    document.body.classList.remove('dark');
}

window.onload = function () {
    let saved = localStorage.getItem('theme');
    if (saved === 'dark') {
        document.body.classList.add('dark');
    }
};

```

4. Output:



5. Result/Conclusion:

The Web Storage API was implemented to persist user preferences such as theme settings using `localStorage.setItem()`, `getItem()` and `removeItem()`, allowing a Dark/Light mode toggle to remain consistent across browser reloads. This reinforced the use of `localStorage` and `sessionStorage` for client-side data persistence in JavaScript.